

The Ninomiya-Victoir Weak Approximation Scheme

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1 Introduction

The **Ninomiya-Victoir (NV) scheme**, introduced in 2008, is a high-order numerical method designed for the **weak approximation** of Stochastic Differential Equations (SDEs). Unlike strong schemes that aim to track individual sample paths, weak schemes focus on calculating expectations of the form $E[f(X_T)]$, which is the primary requirement for derivative pricing in finance.

The NV scheme achieves a **weak convergence order of 2**, significantly outperforming the standard Euler-Maruyama scheme (order 1).

2 The General Framework

Consider a d -dimensional SDE in Stratonovich form:

$$dX_t = V_0(X_t)dt + \sum_{j=1}^m V_j(X_t) \circ dW_t^j \quad (1)$$

where V_0 is the Stratonovich drift and $V_1, \dots, V_m$ are vector fields. The symbol $\circ$ denotes the Stratonovich integral, which allows for the use of classical calculus rules (such as the chain rule).

2.1 The Algorithm

For a time step h , the scheme approximates the transition $X_k \rightarrow X_{k+1}$ by composing the flows of the vector fields. Let $\exp(tV)(x)$ denote the solution at time t of the ODE $\frac{dy}{dt} = V(y)$ starting at $y(0) = x$.

At each step, we draw a Bernoulli random variable $\Lambda_k \in \{1, -1\}$ with $P(\Lambda_k = 1) = 1/2$.

- **If $\Lambda_k = 1$:**

$$X_{k+1} = \exp\left(\frac{h}{2}V_0\right) \circ \exp(\Delta W^m V_m) \circ \dots \circ \exp(\Delta W^1 V_1) \circ \exp\left(\frac{h}{2}V_0\right)(X_k)$$

- **If $\Lambda_k = -1$:**

$$X_{k+1} = \exp\left(\frac{h}{2}V_0\right) \circ \exp(\Delta W^1 V_1) \circ \dots \circ \exp(\Delta W^m V_m) \circ \exp\left(\frac{h}{2}V_0\right)(X_k)$$

3 Conversion to Stratonovich Form

To apply the NV scheme to an Itô SDE $dX_t = \mu(X_t)dt + \sigma(X_t)dW_t$, we must convert the drift. The Stratonovich drift V_0 is defined as:

$$V_0 = \mu - \frac{1}{2} \sum_{j=1}^m \sum_{k=1}^d \sigma_{kj} \frac{\partial \sigma_{\cdot j}}{\partial x_k} \quad (2)$$

This correction ensures that the deterministic "splitting" steps do not introduce bias into the stochastic process.

4 Application: The Heston Model

The Heston model variance process in Itô form is given by:

$$dv_t = \kappa(\theta - v_t)dt + \sigma\sqrt{v_t}dW_t \quad (3)$$

Using the correction term $-\frac{1}{2}(\sigma\sqrt{v})(\frac{\sigma}{2\sqrt{v}}) = -\frac{\sigma^2}{4}$, the Stratonovich form is:

$$dv_t = \left[\kappa(\theta - v_t) - \frac{\sigma^2}{4} \right] dt + \sigma\sqrt{v_t} \circ dW_t \quad (4)$$

4.1 Exact Flow Solutions

The power of the NV scheme for Heston lies in the fact that the flows can be solved analytically:

1. **Diffusion Flow** ($V_1 \circ dW_t$): Solving $dv = \sigma\sqrt{v} \circ dW$ via the chain rule yields:

$$\sqrt{v(t)} = \sqrt{v(0)} + \frac{\sigma}{2}\Delta W \implies v(t) = \left(\sqrt{v(0)} + \frac{\sigma}{2}\Delta W \right)^2$$

2. **Drift Flow** ($V_0 dt$): The linear ODE $dv = [\kappa(\theta - v) - \frac{\sigma^2}{4}]dt$ yields:

$$v(t) = e^{-\kappa t}v(0) + (1 - e^{-\kappa t}) \left(\theta - \frac{\sigma^2}{4\kappa} \right)$$

5 Conclusion

The Ninomiya-Victoir scheme provides a robust, second-order weak approximation that preserves the positivity of variance in models like Heston. By splitting the SDE into analytically solvable Stratonovich flows, it offers high precision in Monte Carlo simulations with significantly fewer time steps than Euler-based methods.